PUBLICATION CHECKPOINT / v0.3.1

Relativistic ell=1 boson stars: background and radial-mode certification

A reproducible numerical evidence report for the equilibrium sequence, radius-definition audit, radial stability transition, ground mode, and first overtone.

RADIAL GATE B: PASS	GATE A: IN PROGRESS	FULL GATE B: IN PROGRESS

Headline result

sigma0^2 = 2.4004311443e-4 at a1^0 = 0.08, with a conservative deterministic-systematics bound of 7.70e-9 in absolute sigma0^2. The domain-converged first overtone is sigma1 = 0.0907036982.

Prepared 22 August 2026

Tagged numerical release: radial-v0.3.1-hardened

Scope: certified background collocation and relativistic radial ell=1 perturbations. Ordinary ell=0 and new nonradial QNMs are not included.

Abstract

We present a publication-style certification of static ell=1 boson-star backgrounds and their relativistic radial pulsations. The equilibrium sequence reproduces the published frequency at a1^0=0.08 and the maximum-mass configuration. A dedicated radius audit finds R99 = 10.18 for the maximum-mass grid model, while the published value 12.75 closely tracks R999 = 12.70; this discrepancy remains unresolved and prevents Gate A from closing. Radial modes are computed with two well-conditioned discretizations: a nonlinear global boundary-value problem and an independently transcribed, equilibrated Chebyshev generalized eigenproblem. The methods agree in eigenvalue and eigenfunction shape, reproduce the stability sign change, and identify the one-node first overtone. Conservative deterministic uncertainty budgets are reported separately for both low modes. The radial portion of Gate B passes; full Gate B awaits an ordinary ell=0 boson-star QNM benchmark.

Evidence chain

equations	**boundary data**	**two discretizations**	**convergence**	**physics**	**gate**

Claims and non-claims

The report certifies the radial benchmark as a numerical result. It does not close the independent-background requirement, resolve the R99/R999 literature discrepancy, validate an ordinary boson-star QNM solver, or present any new nonradial mode. Algebraic pencil residuals are treated as discretization diagnostics rather than continuum error estimates. The spectral condition number is reported because low algebraic residual alone does not establish a well-conditioned physical eigenvalue.

Report map

Sections 1-2 define conventions and background equations. Section 3 gives the equilibrium evidence and radius audit. Sections 4-5 derive the radial problem and document both numerical formulations. Section 6 assembles the benchmark, convergence, eigenfunction, and uncertainty evidence. Sections 7-9 state limitations, gate decisions, and the reproducibility manifest. Six figures and eight interpretive tables are supported by versioned CSV and JSON records.

1. Scope, conventions, and acceptance policy

The project uses geometrized units G = c = hbar = mu = 1 and metric signature (-,+,+,+). The background line element is written in areal radius as ds^2 = -alpha(r)^2 dt^2 + gamma(r)^2 dr^2 + r^2 dOmega^2, with gamma^2 = [1 - 2M(r)/r]^-1. Perturbations vary as exp(-i sigma t), so Im(sigma) < 0 denotes damping and Im(sigma) > 0 growth. These choices are frozen in the repository and every comparison is converted into this normalization.

Table 1. Frozen units, signs, normalizations, and acceptance rules.

Quantity	Project convention	Consequence
Units	G = c = hbar = mu = 1	Frequencies and radii are dimensionless
Metric	(-,+,+,+), areal radius	gamma^2 = (1-2M/r)^-1
Matter	2 ell+1 equal-mass complex fields	kappa_ell = 2 ell+1
Center amplitude	psi/r^ell -> a_ell^0/kappa_ell	Reproduces the published Table II normalization
Time dependence	exp(-i sigma t)	Im(sigma)<0 is damped
Radius	M(R99)=0.99 M_T	R999 is stored separately
Certification	domain + resolution + method + residual	Solver success alone cannot pass a gate

Numerical acceptance

A headline mode must survive outer-domain variation, resolution variation, background-representation checks, direct boundary or operator residual diagnostics, and an independent discretization. Mode identity is checked by resolved nodes and normalized physical-eigenfunction overlap. Deterministic shifts are summed conservatively for the headline uncertainty; quadrature is retained only as a secondary descriptive diagnostic.

```
accepted radial mode = converged BVP + equilibrated spectral pencil + shape agreement +
physical boundary data + conservative uncertainty
```

2. Relativistic ell=1 backgrounds

2.1 Field equations

For kappa = 2 ell+1, the static Einstein-Klein-Gordon reduction evolves the Misner-Sharp mass M, lapse alpha, radial field psi, and its derivative. In compact notation the implemented equations are

```
M' = (kappa r^2/2)[psi'^2/gamma^2 + (omega^2/alpha^2 + 1 + ell(ell+1)/r^2)psi^2]
alpha'/alpha = gamma^2{M/r^2 + (kappa r/2)[psi'^2/gamma^2 + (omega^2/alpha^2 - 1 -
ell(ell+1)/r^2)psi^2]}
psi'' + (2/r + alpha'/alpha - gamma'/gamma)psi' + gamma^2[omega^2/alpha^2 - 1 -
ell(ell+1)/r^2]psi = 0.
```

Regularity fixes psi = (a_ell^0/kappa) r^ell + O(r^(ell+2)) and M = O(r^(2ell+1)). The outer boundary imposes the Schwarzschild lapse and a massive decaying Robin tail through first order in 1/r. The eigenfrequency omega is constrained to 0 < omega < 1 by an unconstrained logistic parameter.

2.2 Collocation and continuation

The nonlinear background boundary-value problem is solved by adaptive collocation on a geometric radial mesh. Continuation begins at the validated a1^0=0.08 anchor and advances along the nodeless branch. Stored profiles contain M, alpha, gamma, psi, psi', global charges and radii, tail diagnostics, and solver status. Gate A nevertheless remains open because a genuinely independent background shooting formulation has not yet reproduced the profiles.

3. Background sequence and radius audit

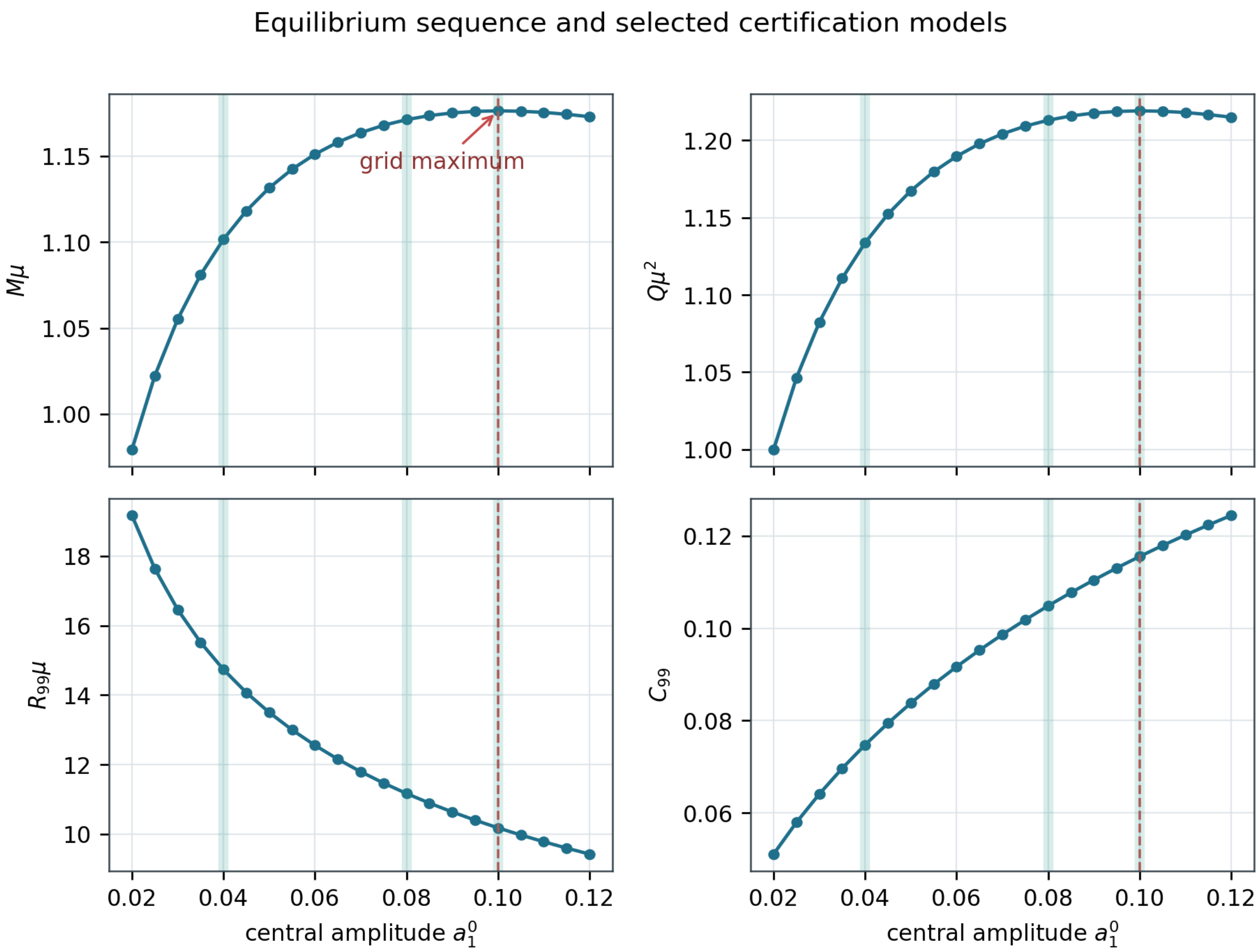

Figure 1. The 21-model equilibrium sequence from a1^0=0.02 to 0.12. The vertical dashed line marks the maximum-mass grid point; translucent bands identify representative profiles used throughout this report. The sequence data are stored in data/ell1_sequence.csv.

3.1 Representative models

Table 2. Representative equilibrium models used for plots and perturbation checks.

Model	a1^0	omega	M_T	Q	R99	C99
dilute	0.040	0.89642580	1.1016307	1.1337079	14.7365	0.07476
benchmark	0.080	0.85189740	1.1712712	1.2129669	11.1673	0.10488
maximum grid	0.100	0.83567215	1.1763866	1.2190106	10.1782	0.11558

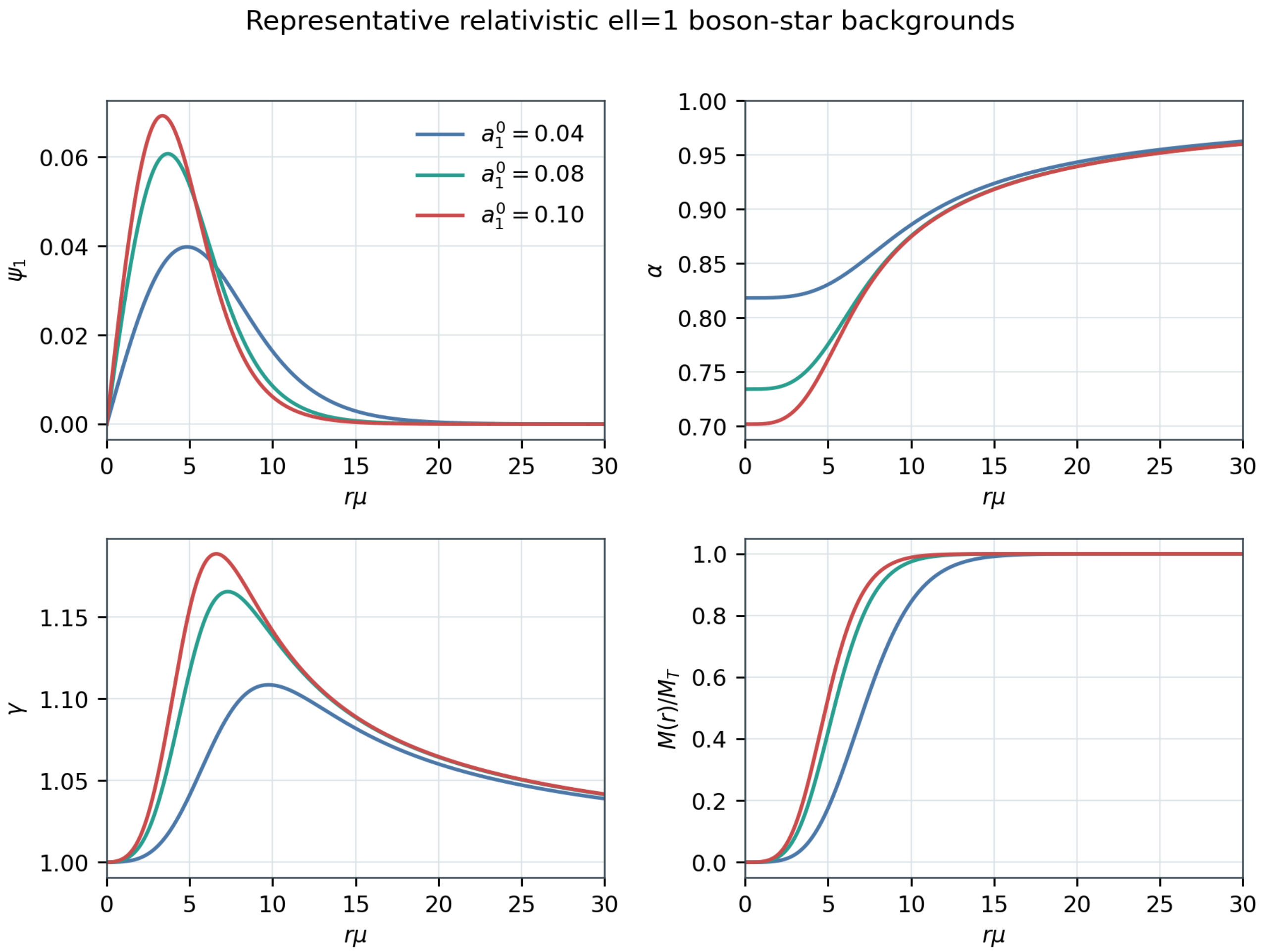

Figure 2. Scalar, lapse, radial-metric, and enclosed-mass profiles for dilute, benchmark, and maximum-mass-grid configurations. Increasing central amplitude compacts the matter distribution and deepens the relativistic lapse.

3.2 Published benchmarks

Table 3. Published-versus-computed background and radial benchmarks from Alcubierre et al. (2021).

Benchmark	Published	Computed	Assessment
a1^0=0.08 frequency	omega = 0.8519	omega = 0.8518974	agrees at quoted precision
maximum model frequency	omega about 0.836	omega = 0.8356722	consistent
maximum model mass	M about 1.18	M = 1.1763866	consistent
radial ground mode	sigma0^2 = 2.40e-4	2.4004311443e-04	independently certified
first radial overtone	sigma1 = 0.0907	0.0907036982	agrees beyond quoted precision

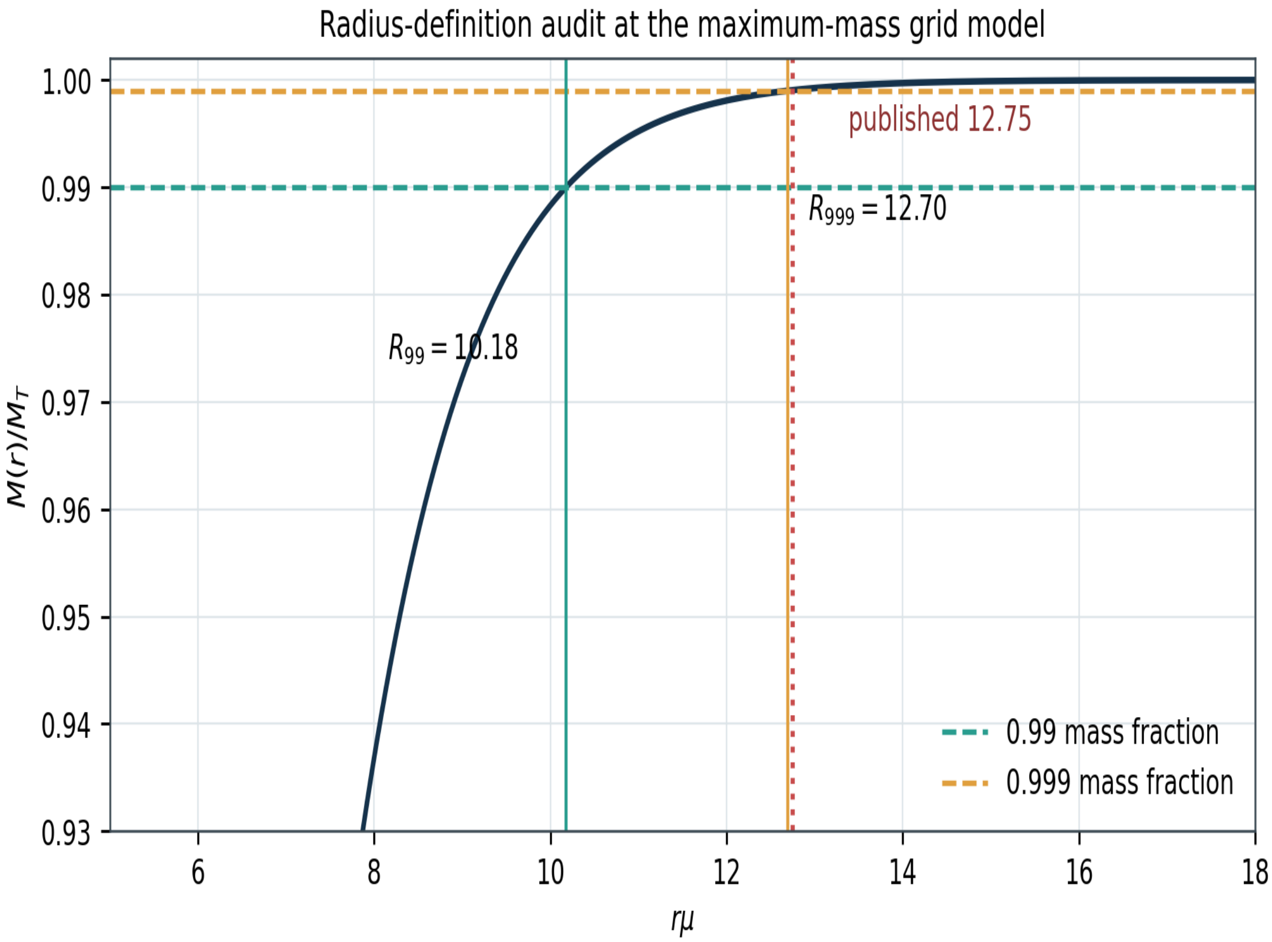

Figure 3. Enclosed mass fraction for the maximum-mass grid model. The literal 99 percent radius is 10.18, whereas the published 12.75 value lies near the independently computed 99.9 percent radius 12.70. The plot supports an unresolved labeling or numerical discrepancy, not a claim of published error.

Table 4. Radius-definition audit at a1^0=0.10.

Definition	Radius	Mass fraction / note
M(R)=0.99 M_T	10.178177	literal R99
M(R)=0.999 M_T	12.697043	literal R999
Published value	12.75	computed enclosed fraction = 0.9990485
Proper-energy 99 percent	10.173811	cross-check
Noether-charge 99 percent	10.135752	cross-check

3.3 Outer-domain convergence

Table 5. Background outer-domain convergence for a1^0=0.08 at fixed collocation tolerance 1e-7.

r_max	omega	M_T	R99	tail residual
60	0.851897404302	1.171271161072	11.16752149	9.04e-14
80	0.851897404290	1.171271161113	11.16733443	8.94e-15
100	0.851897404282	1.171271161143	11.16745834	9.51e-16

The frequency and mass are stable across the tested outer domains to substantially more digits than required by the published benchmark. This is strong same-formulation convergence evidence, but it is not the missing independent background method.

4. Relativistic radial perturbation problem

4.1 Variables and eigenvalue

The radial Appendix system is implemented in delta-varphi_11 and delta-L = delta-lambda/(2 kappa psi_1^2), with sigma^2 as the eigenvalue. Using sigma^2 permits stable and unstable modes to share a real-valued code path. The physical comparison fields are psi delta-varphi_11 and delta-lambda. At the regular center, the normalization and leading series determine five boundary conditions for four first-order fields plus the unknown eigenvalue.

```
y = (delta-varphi, delta-varphi', delta-L, delta-L')
y' = F(r, y; sigma^2),    A x = sigma^2 B x
```

4.2 Physical outer conditions

The global BVP directly imposes the two published finite-radius physical conditions F1=F2=0. For the spectral pencil they are theoretically equivalent to Dirichlet conditions on both numerical perturbation variables because the background scalar and omega are nonzero at the finite outer boundary. Endpoint residuals demonstrate that the imposed conditions are met; domain, resolution, representation, and cross-method changes provide the independent numerical evidence.

4.3 Why affine outward shooting was rejected

The original outward-basis experiment subtracted exponentially amplified solutions. It could reproduce an interior profile while producing misleading normalized boundary residuals through catastrophic cancellation. It remains in the repository only as deprecated historical code. No result or uncertainty in this report uses it for certification.

5. Two radial discretizations

5.1 Nonlinear global BVP

SciPy's adaptive collocation solver treats sigma^2 as a free parameter and enforces the regular-center and physical outer equations simultaneously. An outward IVP supplies only the Newton starting profile. Diagnostics distinguish the maximum SciPy interval RMS relative residual from a separately sampled maximum pointwise relative ODE residual. This terminology is retained in the machine-readable table.

5.2 Independent generalized eigenproblem

The interval [epsilon,r_max] is discretized at Chebyshev-Lobatto points. A separately transcribed coefficient module assembles the linear pencil A x = sigma^2 B x without calling the BVP right-hand side or center coefficient. Tests compare the assembled interior operator and all center/outer rows against direct evaluations on arbitrary smooth fields. The pencil is equilibrated by positive row and column scalings before the dense generalized eigensolve.

```
A_s = D_r A D_c,    B_s = D_r B D_c,    A_s v = sigma^2 B_s v
```

5.3 Root filtering and mode tracking

Finite eigenvalues are retained only when their imaginary contamination lies below the numerical tolerance and their real value falls inside the requested search interval. Physical fields are reconstructed before classification. Resolved interior zeros are located by shape-preserving interpolation and bracketing with only a floating-point noise floor; the former fixed 1e-4 amplitude cutoff is not used. Across resolutions, normalized L2 eigenfunction overlap supplements frequency and nodal continuity.

5.4 Background smoothness policy

The certification solver requires a C1 cubic-Hermite representation of u=psi/r^ell built from stored psi and psi'. PCHIP is rejected by the public spectral API because a small matrix residual can coexist with a biased continuum eigenvalue when coefficient derivatives are insufficiently smooth. PCHIP is retained only as a local-BVP representation variation.

6. Radial spectrum and certification

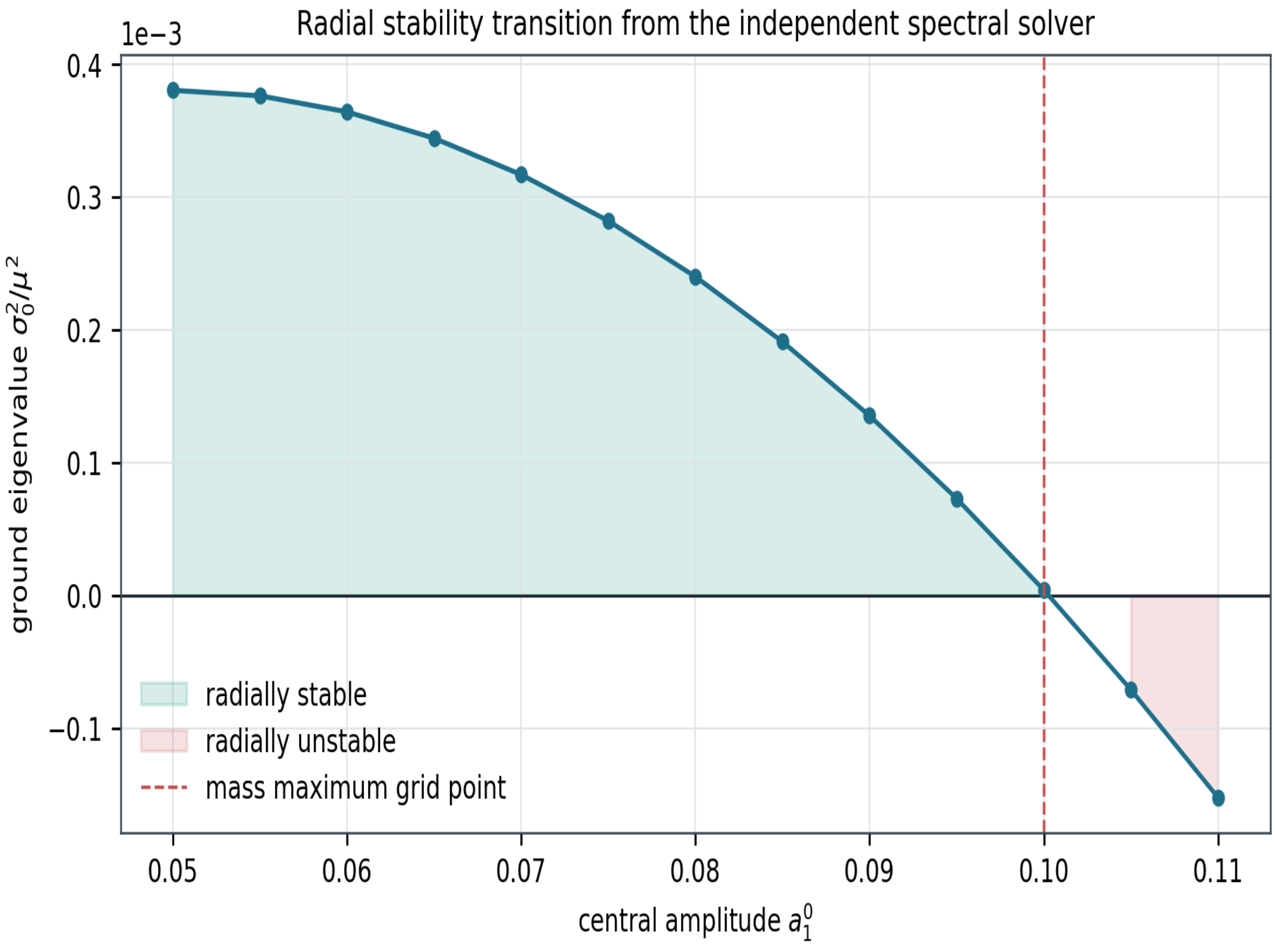

Figure 4. Ground radial eigenvalue across the central-amplitude sequence. Positive sigma0^2 denotes oscillatory stability; the sign becomes negative beyond the maximum-mass neighborhood, reproducing the turning-point stability transition.

6.1 Low-mode frequencies

Table 6. Published-versus-computed relativistic radial frequencies at a1^0=0.08.

Mode	Method/domain	sigma^2	sigma	nodes	Published
ground	BVP, r_max=40	2.400431144339e-04	0.0154933248	0	sigma0^2=2.40e-4
first overtone	BVP, r_max=60	8.227160870356e-03	0.0907036982	1	sigma1=0.0907

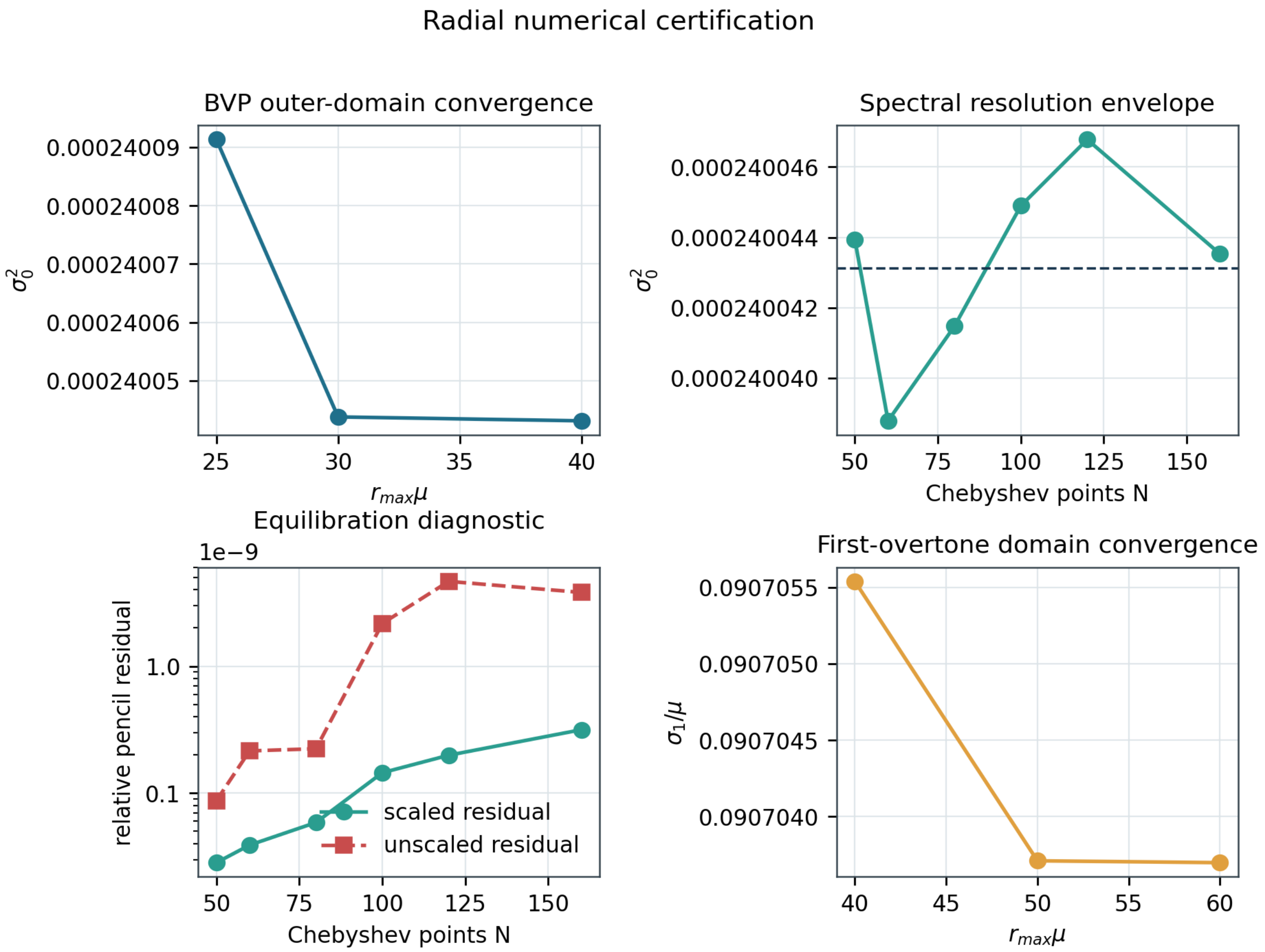

Figure 5. Four complementary convergence diagnostics: BVP outer-domain convergence, the broad N=50-160 spectral envelope, scaled versus unscaled algebraic residuals, and the stronger outer-domain sensitivity of the first overtone. Oscillatory spectral drift motivates an envelope rather than a monotonic extrapolation.

6.2 Cross-method comparison

Table 7. BVP-spectral comparison at common finite domains. The overtone condition number quantifies the harder large-domain pencil.

Mode/domain	BVP sigma^2	Spectral sigma^2	absolute difference	condition number
ground, r_max=40, N=80	2.400431144339e-04	2.400414709824e-04	1.643e-09	4.931e+01
overtone, r_max=60, N=160	8.227160870356e-03	8.227197303978e-03	3.643e-08	4.198e+07

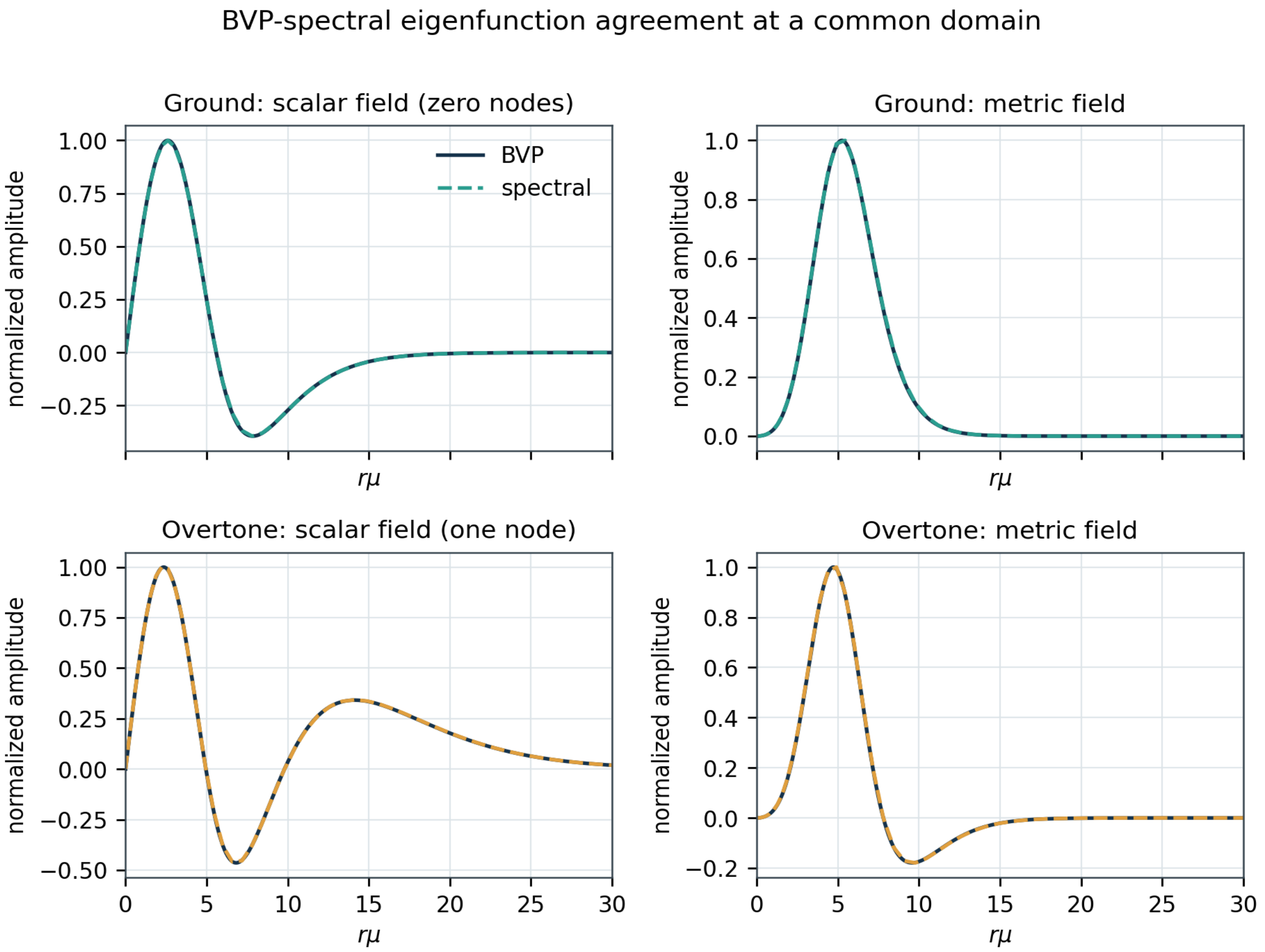

Figure 6. Physical scalar and metric eigenfunctions for the ground mode and first overtone at a common r_max=40 domain. Curves are independently normalized. Their near-coincidence verifies mode shape, while the overtone panels visibly contain one resolved interior node.

6.3 Eigenfunction agreement

At the common domain, the normalized L2 overlaps are 0.999999273 for the physical scalar perturbation and 0.999995403 for delta-lambda. This shape comparison is stronger evidence than eigenvalue agreement alone because it probes the reconstructed physical fields across the radial interval.

7. Deterministic numerical uncertainty

Table 8. Conservative numerical-systematics budgets. Components are not assumed statistically independent.

Mode/component	absolute shift in sigma^2	Interpretation
ground: outer domain r30 to r40	6.741e-10	systematic variation
ground: bvp to spectral n80	1.643e-09	systematic variation
ground: spectral resolution max displacement from bvp n50 to n160	4.334e-09	systematic variation
ground: hermite to pchip	1.051e-09	systematic variation
GROUND CONSERVATIVE SUM	7.702e-09	headline bound
overtone: outer domain r40 to r60	3.340e-07	systematic variation
overtone: bvp to spectral at r60 n160	3.643e-08	systematic variation
overtone: spectral resolution max displacement from r40 bvp n50 to n160	2.732e-07	systematic variation
OVERTONE CONSERVATIVE SUM	6.436e-07	headline bound

Recommended reporting

For the ground mode, a defensible rounded result is sigma0^2 = (2.40043 +/- 0.00008)e-4, where the uncertainty is a deterministic numerical-systematics bound. For the first overtone, sigma1 = 0.09070370 is the domain-converged reference, but the substantially larger sigma1^2 bound should accompany any precision claim. Neither uncertainty is a statistical standard deviation.

The ground budget includes BVP domain, BVP-spectral method, broad spectral-resolution, and Hermite-versus-PCHIP local-BVP representation changes. The overtone budget separately includes its r_max=40 to 60 domain shift, the r_max=60 BVP-spectral comparison, and the maximum displacement of the r_max=40 spectral resolution sequence from the common-domain BVP.

8. Limitations, gate decision, and next milestone

8.1 What is certified

The relativistic radial ell=1 benchmark passes. Two independent discretizations reproduce the ground eigenvalue, stability sign crossing, first overtone, and physical eigenfunction shapes. Direct matrix-operator tests independently validate the spectral coefficient transcription. The embedded release history and runtime provenance make the result reproducible.

8.2 What remains open

Gate A remains in progress because the background collocation profiles still lack an independent shooting reproduction and the radius discrepancy is unresolved. Full Gate B remains in progress because the ordinary ell=0 boson-star QNM benchmark has not been implemented. A smooth-background mapped multidomain architecture, explicit branch handling for massive scalar sidebands, constraint and gauge diagnostics, and a time-domain or other independent physical validation are required before new nonradial spectra can be frozen for publication.

Gate	Decision	Evidence / blocker
A: background	IN PROGRESS	Collocation converged; independent shooting and R99 discrepancy remain
B: radial portion	PASS	BVP + independent equilibrated spectral formulation + shape agreement
B: full	IN PROGRESS	Ordinary ell=0 QNM benchmark remains
Production nonradial	NOT CERTIFIED	Multidomain QNM system, constraints, branches, and independent validation remain

8.3 Next hard milestone

The next miniature publication report should treat ordinary ell=0 boson-star QNMs with the same evidence chain: asymptotics and branch conventions, a frequency-domain solver, three-resolution or contour-based root filtering, published scalar-led and gravitational-led benchmarks, eigenfunctions, a time-domain or second-method cross-check, and an explicit uncertainty/provenance table. No new ell=1 nonradial frequency should be promoted to a paper claim before that benchmark passes.

9. Reproducibility manifest

Item	Value
Report source commit	41c608bd49e6ae8664a96aa2b9ff243821b12941
Report source tree	3141d64d2bdcee7905f2d4392a2b7129d3199a3e
Certified numerical implementation	a5f1385069ff4fa0886e5584464f0b18232e4c3a
Release tag	radial-v0.3.1-hardened
Python	3.14.3
NumPy / SciPy	2.4.3 / 1.17.1
Platform	Windows-11-10.0.26200-SP0
Top-level pin SHA-256	f419489ab39b3955cc143f63108082defee9f7d7f87897a92470d034104f7e75
Background sequence	data/ell1_sequence.csv
Radial certification	data/radial_benchmarks.csv (27 rows)
Ground uncertainty	data/radial_uncertainty.json
Overtone uncertainty	data/radial_overtone_uncertainty.json
Report stability curve	data/report_radial_stability.csv (13 regenerated models)
Automated suite	20 tests

Regeneration commands

```
python -m radial.diagnostics --output data/radial_benchmarks.csv --uncertainty-output
data/radial_uncertainty.json --overtone-uncertainty-output
data/radial_overtone_uncertainty.json
python paper/build_certification_report.py
python -m pytest -q
```

Data and code statement

All plotted radial certification values and uncertainty components are stored in machine-readable repository files. Figure source PDFs and high-resolution raster previews are generated in the temporary report asset directory; the final report is emitted under output/pdf. The tagged Git bundle in the presentation release records the complete numerical history.

References

[1] M. Alcubierre et al., On the linear stability of ell-boson stars with respect to radial perturbations, Phys. Rev. D 103, 124046 (2021), arXiv:2103.15012.

[2] M. Alcubierre et al., L-boson stars, Class. Quantum Grav. 35, 19LT01 (2018), arXiv:1805.11488.

[3] D. Batic, D. Dutykh, and M. E. Sukaiti, Quasinormal modes of the Kazakov-Solodukhin quantum-corrected black hole: a spectral analysis, Eur. Phys. J. C 86, 847 (2026), doi:10.1140/epjc/s10052-026-16102-3. Used as a structural exemplar for evidence-rich spectral reporting, not as a source of boson-star physics.

[4] SciPy documentation for solve_bvp and scipy.linalg.eig, runtime version recorded in the manifest.

Appendix A. Diagnostic definitions

SciPy interval RMS relative residual: the maximum entry of solve_bvp.rms_residuals; this is not a maximum pointwise defect. Dense pointwise relative residual: max |y'_collocation - F(r,y)|/(1+|F|) on a 3000-point geometric audit grid. Scaled generalized residual: ||A_s v - sigma^2 B_s v||/[||A_s v|| + |sigma^2| ||B_s v||]. Unscaled generalized residual: the same expression after transforming the right eigenvector back to the original pencil. Left/right condition number: ||w|| ||v||/|w^H B_s v| for the equilibrated pencil. These algebraic quantities diagnose the discrete solve; continuum uncertainty comes from domain, resolution, representation, and method changes.

Appendix B. Release checklist

Check	Status
Portable tagged source archive	complete in radial-v0.3.1-hardened release
Full Git history bundle	verified
Runtime and pin provenance	recorded in CSV/JSON and this report
BVP and spectral low modes	reproduced
Independent spectral coefficients	direct operator identity tests pass
Conservative ground uncertainty	7.70e-9 absolute sigma0^2
Separate overtone uncertainty	6.44e-7 absolute sigma1^2
Ordinary ell=0 QNM	not implemented
New nonradial spectrum	not implemented